Holy Cross College

Semester 1 Examination 2019

Question/Answer Booklet

Year 10 Science (General)

Physical Science

Please place your student identification label in this box

Student Name SOLUTIONS

Student's Teacher ______________________

Time allowed for this paper

Reading time before commencing work: 10 minutes
Working time for paper: 2 hours

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	7	7	10	7	
Section Two: Short Answer	6	6	20	27	
Section Three: Comprehension					
				34	

Instructions to candidates

1. The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer Booklet.
3. Working or reasoning should be clearly shown when calculating or estimating answers.
4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
   - Fill in the number of the question(s) that you are continuing to answer at the top of the page.

SECTION 1 MULTIPLE CHOICE (7 marks)

Choose the best answer for each question in the Multiple Choice Question booklet.

Write your answer in the box to the right.

MULTIPLE CHOICE ANSWERS	
1	B
2	A
3	D
4	A
5	C
6	A
7	D

1. Calculate the ***speed*** of a bus that travels between two traffic lights 145 metres apart in 20.0 seconds.

   (a) 7.00 m/s
   (b) 7.25 m/s
   (c) 6.25 m/s
   (d) 7.50 m/s

2. ***How long*** would a car travelling at a speed of 60 km/h take to cover a distance of 90 km?

   (a) 1½ hours
   (b) 2 ½ hours
   (c) 3 ½ hours
   (d) 4 ½ hours

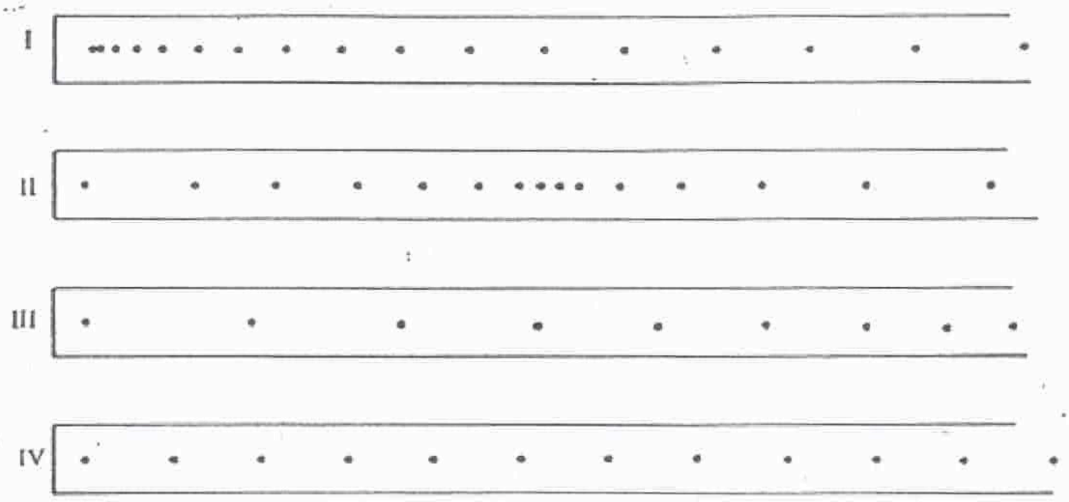
3. An object moves with a speed of 8 m/s for 22 seconds. What ***distance*** has it travelled?

   (a) 0.364 m
   (b) 2.75 m
   (c) 30 m
   (d) 176 m

4. Speed is a measure of the rate:

   (a) at which distance is covered.
   (b) at which speed increases.
   (c) of change in force.
   (d) of change in acceleration.

The following diagrams represent the marks made on four separate tapes by the timing device as the cart moves.



5. Which one of the diagrams best indicates that the cart was ***slowing down*** throughout the recorded time interval?

   (a) I (b) II (c) III (d) IV

6. Which one of the diagrams best indicates that the cart ***started from rest and accelerated***?

   (a) I (b) II (c) III (d) IV

7. Which one of the diagrams best indicates that the cart was ***travelling at constant speed***?

   (a) I (b) II (c) III (d) IV

SECTION 2 SHORT ANSWER (27 marks)

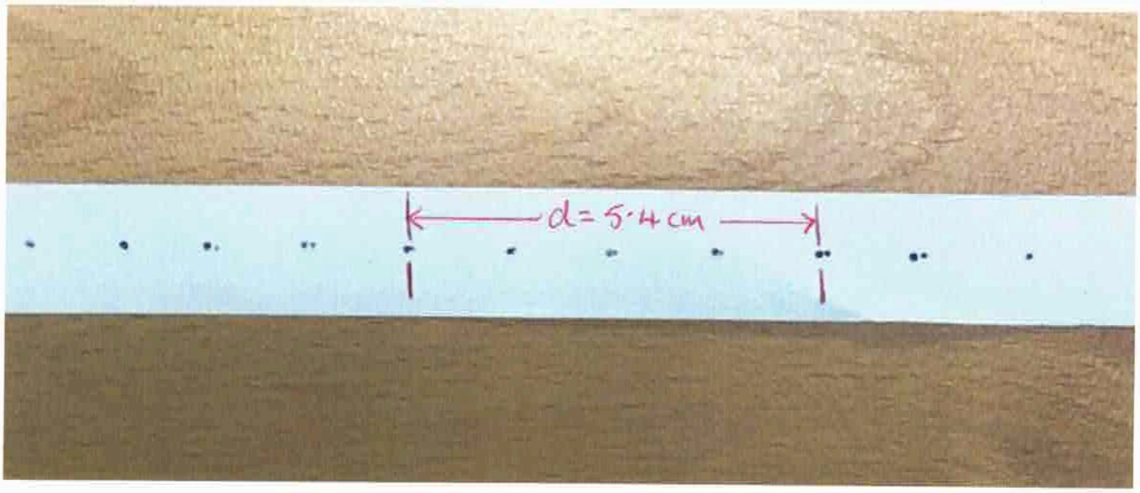
Attempt all questions. Write your answers in the spaces provided.

1. Complete the following table. Include the correct units. (3 marks)

Distance travelled	Time taken	Average speed
200 m	25 s	8 m/s
375 m	25 s	15 m/s
2860 m	130 s	22 m/s

[1 mark each]

2. The following ticker tape shows the motion of an object in the Science laboratory. (1 dot spacing = 0.02 seconds)



(a) Describe the motion of the object. (1 mark)

- constant speed. (1)

(b) Calculate the average speed shown for the five dots indicated. (3 marks)

sp =

d = 0·054 m

t = 0·08 s

$sp = \frac{d}{t}$ (1)

$= \frac{0.054}{0.08}$ (1)

$= \underline{0.675 \text{ m/s}}$ (1)

3. Tim and Peter are good friends who live in the Rivervale area. Tim was planning to walk from his place to Peter's and the map below shows their respective houses. Use the map to do the following.

   (a) ***Trace on the map*** the way you would walk to Peter's house. Using the scale, calculate (approximately) the ***distance*** you would walk. (2 marks)

   d = 965 m (± 30 m) (2)

   (b) Use the scale to determine the ***displacement*** and ***bearing*** from Tim's house to Peter's house. Remember to draw the compass directions on the map. (3 marks)

   s = 680 m (± 30 m) bearing 95° (± 5°)
   (1) (1)

   (c) Explain the key difference between ***distance*** and ***displacement*** in Physics. (2 marks)

   DISTANCE: amount of movement in any direction (1)

   DISPLACEMENT: straight line motion in a particular direction (1)

4. You are flying from Perth to Singapore, a distance of 3912 kilometres. It takes 5.5 hours to make the journey. What is the average speed of the plane? (3 marks)

$sp = ?$

$d = 3912 \text{ km}$

$t = 5.5 \text{ hrs}$

$sp = \frac{d}{t}$ (1)

$= \frac{3912}{5.5}$ (1)

$= \underline{711 \text{ km/h}}$ (1)

5. A professional cyclist in a race averages 43.62 km/h for 0.583 hours in a time trial. What was the length of the stage? (3 marks)

$sp = 43.62 \text{ km/h}$

$d = ?$

$t = 0.583 \text{ hrs}$

$d = sp \times t$ (1)

$= 43.62 \times 0.583$ (1)

$= \underline{25.4 \text{ km}}$ (1)

6. The Spanish MotoGP was held recently at Circuito de Jerez-Angel Nieto track, which is 4.428 km long. The race was held over 25 laps, with Marc Márquez winning with an average speed of 161.2 km/h.

(a) What is the average speed of Márquez in metres/second (m/s)? (2 marks)

$$sp = \frac{161 \cdot 2}{3 \cdot 6} \quad (1)$$

$$= \underline{44 \cdot 78 \text{ m/s}} \quad (1)$$

(b) What was the distance of the race in metres? (2 marks)

$$d = 4 \cdot 428 \times 25 \quad (1)$$

$$= 110 \cdot 7 \text{ km}$$

$$= \underline{110{,}700 \text{ m}} \quad (1)$$

(c) What time (in seconds) did Márquez take to complete the race? (3 marks)

$sp = 44 \cdot 78$ m/s

$d = 110{,}700$ m

$t = ?$

$$t = \frac{d}{sp} \quad (1)$$

$$= \frac{110{,}700}{44 \cdot 78} \quad (1)$$

$$= \underline{2472 \text{ s}} \quad (1)$$